

# MAT 257 Analysis II

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2026 Fall

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# Chapter 1

## Lecture Notes

### 1.1 Continuity

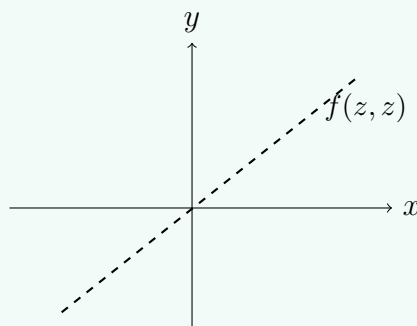
#### Wrong Concept 1.1: Continuity by components

A scalar function  $f(x_1, \dots, x_n)$  is continuous if it is continuous as a function of each  $x_i$ , regarding the other variables as constant.

More generally, a vector valued  $f(x_1, \dots, x_n)$  is continuous if all  $f_j(x_1, \dots, x_n)$  are continuous.

**Example 1.1.1** (Separate continuity is not enough)

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$



Restricting to the diagonal,

$$f(z, z) = \frac{z^2}{2z^2} = \frac{1}{2}$$

Thus  $f \circ g$  with  $g(z) = (z, z)$  is not continuous. This is a counterexample to the definition above.

## Wrong Concept 1.2: Limits along lines

For  $(a_1, \dots, a_n)$ ,

$$\lim_{t \rightarrow 0} f(x_1 + ta_1, \dots, x_n + ta_n)$$

exists and doesn't depend on  $a = (a_1, \dots, a_n)$ .

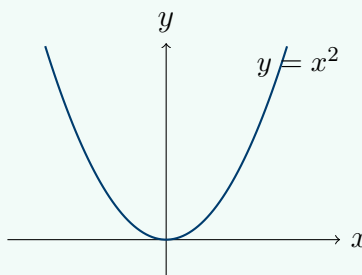
### Example 1.1.2 (A second counterexample)

Let  $y = x^2$  and

$$f(x, y) = \varphi\left(\frac{y}{x^2}\right)$$

for some function  $\varphi(\lambda)$  such that

$$\varphi(1) = c \neq 0, \quad \varphi(0) = 0, \quad \varphi(\lambda) \rightarrow 0 \text{ for } \lambda \rightarrow \infty$$



For example, take

$$\varphi(\lambda) = \frac{\lambda}{1 + \lambda^2}$$

This gives

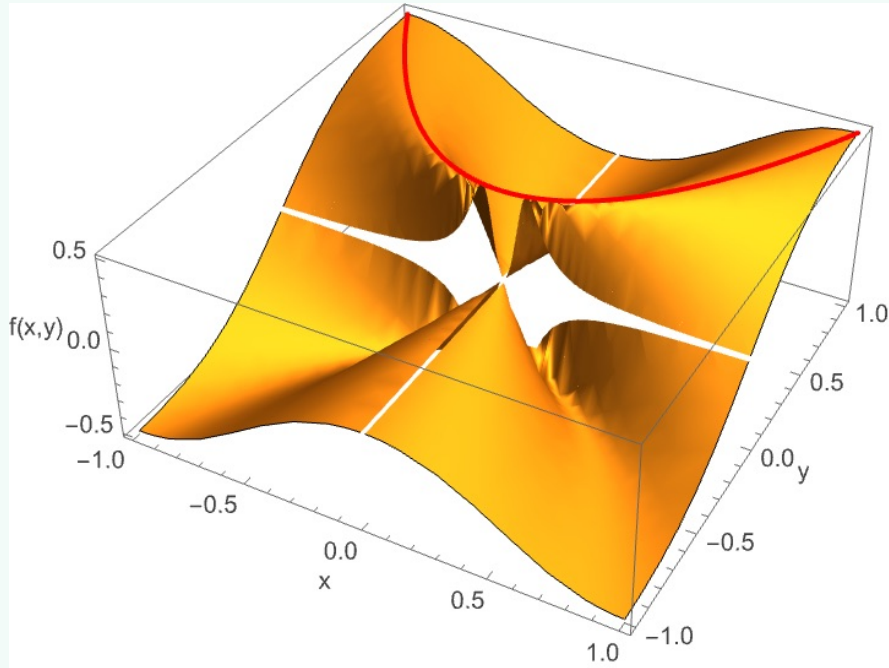
$$\varphi\left(\frac{y}{x^2}\right) = \frac{\frac{y}{x^2}}{1 + \frac{y^2}{x^4}} = \frac{yx^2}{x^4 + y^2}$$

With this choice,

$$f(z, z^2) = \varphi(1) = \frac{1}{2}$$

so  $f(z, z^2) \not\rightarrow 0$  for  $z \rightarrow 0$ . But

$$f(t, t) = \frac{t^3}{t^4 + t^2} = \frac{t}{t^2 + 1} \rightarrow 0 \quad \text{for } t \rightarrow 0$$



### Definition 1.1.1: Continuity at a point

Write  $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ . A function  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is continuous at  $x \in \mathbb{R}^n$  if

$$\lim_{h \rightarrow 0} f(x + h) = f(x)$$

We'll have to define "lim".

#### Note:-

Defining "lim" leads to the topology of  $\mathbb{R}^n$ . Norms on  $\mathbb{R}^n$  are introduced in the next section, and the equivalent  $\epsilon$ - $\delta$  formulation of continuity follows from them.

### Definition 1.1.2: Continuity in $\epsilon$ - $\delta$ form

A function  $f : A \rightarrow \mathbb{R}^m$ , where  $A \subseteq \mathbb{R}^n$  is any subset, is continuous at  $x_0 \in A$  if

$$\forall \epsilon > 0 \exists \delta > 0 \text{ s.t. } \forall x \in A, \|x - x_0\| < \delta \implies \|f(x) - f(x_0)\| < \epsilon$$

We say  $f$  is continuous if it is continuous everywhere.

#### Note:-

- 1) We could replace the Euclidean norm  $\|\cdot\|$  by any equivalent norm, according to the previous definition of norm equivalence

$$c\|x\|_1 \leq \|x\|_2 \leq C\|x\|_1$$

- 2) The subset  $A$  is any subset. For example,  $A$  could be a closed or open interval, or the sphere in  $\mathbb{R}^n$ , or anything.
- 3) If  $f : A \rightarrow \mathbb{R}^m$  is continuous and  $A_1 \subseteq A$ , then  $f|_{A_1}$  is continuous. The converse is not

true. This could be easily proven from the definition.

**Example 1.1.3** (A function whose restriction is nicer than the function)

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = \begin{cases} 0 & \text{if } x \text{ rational} \\ 1 & \text{if } x \text{ irrational} \end{cases}$$

is not continuous, but  $f|_{\mathbb{Q}}$  is rational.

**Proposition 1.1.1** Composition of continuous functions

Let  $A \subseteq \mathbb{R}^n$ ,  $B \subseteq \mathbb{R}^m$ , and

$$f : A \rightarrow \mathbb{R}^m, \quad g : B \rightarrow \mathbb{R}^\ell$$

with  $f(A) \subseteq B$ . If  $f$  is continuous at  $x_0$  and  $g$  is continuous at  $f(x_0)$ , then  $g \circ f$  is continuous at  $x_0$ .

In particular, if  $f, g$  are continuous then  $g \circ f$  is continuous.

**Proof:** Let  $\epsilon > 0$  be given. Since  $g$  is continuous at  $f(x_0)$ , there is  $\delta_g > 0$  such that

$$\|y - f(x_0)\| < \delta_g \implies \|g(y) - g(f(x_0))\| < \epsilon \quad (*)$$

Since  $f$  is continuous at  $x_0$ , there is  $\delta > 0$  such that

$$\forall x \in A, \quad \|x - x_0\| < \delta \implies \|f(x) - f(x_0)\| < \delta_g$$

By taking  $y = f(x)$ , we deduce (\*). So

$$\|x - x_0\| < \delta \implies \|g(f(x)) - g(f(x_0))\| < \epsilon$$

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**Proposition 1.1.2** Continuous functions form a vector space

The set of continuous functions  $f : A \rightarrow \mathbb{R}^m$  is a vector space, under the usual addition and scalar multiplication.

**Proposition 1.1.3** Componentwise continuity

A function  $f : A \rightarrow \mathbb{R}^m$  is continuous iff its component functions are continuous, writing

$$f = (f_1, \dots, f_m)$$

**Exercise 1.1.1** Componentwise continuity

Show the proposition above. *Hint:*

$$\|f(x) - f(x_0)\|_\infty = \max \{|f_j(x) - f_j(x_0)| \mid j = 1, \dots, m\}$$

## 1.2 Norms on $\mathbb{R}^n$

### Definition 1.2.1: Norm

Let  $V$  be a finite dimensional vector space over  $\mathbb{R}$ . A norm on  $V$  is a function

$$\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$$

such that

- (a)  $\|v\| = 0$  iff  $v = 0$
- (b)  $\|tv\| = |t|\|v\|$  for all  $t \in \mathbb{R}, v \in V$
- (c)  $\|v + w\| \leq \|v\| + \|w\|$  for all  $v, w \in V$  (triangle inequality)

### Definition 1.2.2: Euclidean norm

The “standard” norm of  $\mathbb{R}^n$  (Euclidean), for  $x = (x_1, \dots, x_n)$ , is

$$\|x\| = \sqrt{x_1^2 + \dots + x_n^2} = \sqrt{x \cdot x}$$

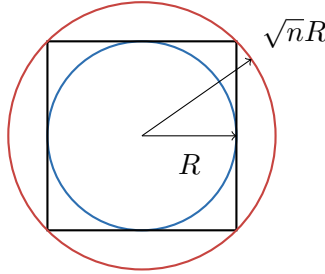
### Definition 1.2.3: Sup-norm

Another important norm on  $\mathbb{R}^n$  is the “sup-norm”:

$$\|x\|_{\infty} = \max \{|x_1|, \dots, |x_n|\}$$

Relationship of the norms:

$$\|x\|_{\infty} \leq \|x\| \leq \sqrt{n} \|x\|_{\infty}$$



### Definition 1.2.4: Equivalent norms

Two norms  $\|\cdot\|_1, \|\cdot\|_2$  on a vector space  $V$  are **equivalent** if

$$\exists c, C > 0 \text{ s.t. } c\|x\|_1 \leq \|x\|_2 \leq C\|x\|_1$$

#### Example 1.2.1 (Euclidean and sup are equivalent)

The Euclidean norm and the sup-norm are equivalent.

#### Proposition 1.2.1 All norms on $\mathbb{R}^n$ are equivalent

Any two norms on  $\mathbb{R}^n$  are equivalent.

(proof later)

**Note:-**

From now on,  $\|\cdot\|$  means the Euclidean norm, unless specified.

**Example 1.2.2** (Norm on an infinite dimensional space)

An example of a norm on an infinite dimensional vector space. Let

$$V = \{\text{continuous functions } f : \mathbb{R} \rightarrow \mathbb{R}\}$$

Then

$$\|f\|_\infty = \sup \{|f(x)| : x \in \mathbb{R}\}$$

is a norm on  $V$ .

## 1.3 Topology of $\mathbb{R}^n$

**Definition 1.3.1:  $\epsilon$ -ball**

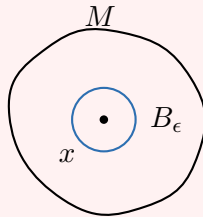
The  $\epsilon$ -ball around  $x \in \mathbb{R}^n$  is

$$B_\epsilon(x) = \{y \in \mathbb{R}^n \mid \|x - y\| < \epsilon\}$$

**Definition 1.3.2: Open subset**

A subset  $M \subseteq \mathbb{R}^n$  is **open** if

$$\forall x \in M \exists \epsilon > 0 \text{ s.t. } B_\epsilon(x) \subseteq M$$



A neighbourhood at  $x$  is an open subset  $U$  with  $x \in U$ .

**Note:-**

$\mathbb{R}^n, \emptyset$  are not open.

**Proposition 1.3.1** Properties of open sets

- (a) Every finite intersection of open sets is open.
- (b) Every union of open sets is open.
- (c)  $\emptyset$  and  $\mathbb{R}^n$  are open.

**Proof:** (c) Vacuously true for  $\emptyset$ , so  $\emptyset$  is open. Also automatic for  $\mathbb{R}^n$ .

- (a) Suppose  $M_1, \dots, M_k \subseteq \mathbb{R}^n$  are open. Let  $x \in M_1 \cap \dots \cap M_k$ . For each  $i = 1, \dots, k$ ,

choose  $\epsilon_i > 0$  such that

$$B_{\epsilon_i}(x) \subseteq M_i$$

Let  $\epsilon = \min\{\epsilon_1, \dots, \epsilon_k\}$ , then

$$B_\epsilon(x) \subseteq \bigcap_{i=1}^k M_i$$

(b) Let  $\{M_\alpha \mid \alpha \in I\}$  be a collection of open sets, where  $I$  is any indexing set. Let  $x \in \bigcup_{\alpha \in I} M_\alpha$ ; choose  $\beta \in I$  with  $x \in M_\beta$ . Let  $\epsilon > 0$  be such that

$$B_\epsilon(x) \subseteq M_\beta \subseteq \bigcup_{\alpha \in I} M_\alpha$$

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### Proposition 1.3.2 Continuity via preimages

A function  $f : A \rightarrow \mathbb{R}^m$  is continuous iff for every  $M \subseteq \mathbb{R}^m$  there exists an open  $V \subseteq \mathbb{R}^n$  such that

$$f^{-1}(M) = A \cap V$$

If  $A$  itself is open,  $f : A \rightarrow \mathbb{R}^m$  is continuous iff for every open  $M \subseteq \mathbb{R}^m$ ,  $f^{-1}(M)$  is open.

**Proof:** “ $\implies$ ” Suppose  $f$  is continuous and  $M \subseteq \mathbb{R}^m$  is open. For every  $x \in f^{-1}(M)$  there is  $\epsilon_x > 0$  such that

$$B_{\epsilon_x}(f(x)) \subseteq M$$

By continuity, we can find  $\delta_x > 0$  such that

$$f(B_{\delta_x}(x) \cap A) \subseteq B_{\epsilon_x}(f(x)) \subseteq M$$

Put

$$V = \bigcup_{x \in f^{-1}(M)} B_{\delta_x}(x)$$

This is open, and

$$V \cap A = \bigcup_{x \in f^{-1}(M)} B_{\delta_x}(x) \cap A \subseteq f^{-1}(M)$$

But also  $A \cap V \supseteq f^{-1}(M)$  since  $B_{\delta_x}(x) \ni x$ .

“ $\impliedby$ ” Suppose the condition holds. To show  $f$  is continuous at  $x_0 \in A$ , let  $\epsilon > 0$  be given. Let

$$M = B_\epsilon(f(x_0))$$

By assumption there is an open  $V$  such that

$$f^{-1}(M) = A \cap V$$

Since  $V$  is open and  $x_0 \in V$ , we can choose  $\delta > 0$  such that  $B_\delta(x_0) \subseteq V$ . Then

$$f^{-1}(B_\epsilon(f(x_0))) = A \cap V \supseteq A \cap B_\delta(x_0)$$

i.e.

$$f(A \cap B_\delta(x_0)) \subseteq B_\epsilon(f(x_0))$$



### Definition 1.3.3: Closed subset

A subset  $A \subseteq \mathbb{R}^n$  is **closed** if its complement

$$A^c = \mathbb{R}^n \setminus A$$

is open.

### Proposition 1.3.3 Properties of closed sets

- (a) Finite unions of closed subsets are closed.
- (b) Arbitrary intersections of closed subsets are closed.
- (c)  $\emptyset, \mathbb{R}^n$  are closed.

**Proof:** Take complements, see exercise.



### Proposition 1.3.4 Interior and closure

Let  $S \subseteq \mathbb{R}^n$  be any subset.

- (a)  $S$  contains a unique largest open subset.
- (b)  $S$  is contained in a unique smallest closed subset.

These subsets are denoted the **interior**  $\text{int}(S)$  and the **closure** of  $S$ ,  $\overline{S}$ .

**Proof:** Define

$$\text{int}(S) = \bigcup_{\substack{U \subseteq S \\ U \text{ open}}} U, \quad \overline{S} = \bigcap_{\substack{A \supseteq S \\ A \text{ closed}}} A$$

The first is open, and the second is closed. Also

$$\text{int}(S) \subseteq S \subseteq \overline{S}$$

By definition, every open set contained in  $S$  is contained in  $\text{int}(S)$ ; every closed set containing  $S$  contains  $\overline{S}$ .



### Example 1.3.1 (Interior and closure of a sequence)

For

$$S = \left\{ \frac{1}{n} \mid n \in \mathbb{N} \right\}$$

we have

$$\text{int}(S) = \emptyset, \quad \overline{S} = S \cup \{0\}$$

## 1.4 Connectedness

### Definition 1.4.1: Connected subset of $\mathbb{R}$

For the real line, a subset  $A \subseteq \mathbb{R}$  is **connected** if for any  $x, y \in A$  with  $x \leq y$ , the interval  $[x, y]$  is contained in  $A$ .

For  $\mathbb{R}^n$  define:

### Definition 1.4.2: Disconnected subset

A subset  $A \subseteq \mathbb{R}^n$  is **disconnected** if there exist disjoint open subsets  $U_1, U_2$  such that

$$A \cap U_1 \neq \emptyset, \quad A \cap U_2 \neq \emptyset, \quad A = (A \cap U_1) \cup (A \cap U_2)$$

It is **connected** otherwise.

### Example 1.4.1 (Punctured line is disconnected)

$A = \mathbb{R} \setminus \{0\}$  is disconnected, since

$$A \cap (-\infty, 0) \neq \emptyset, \quad A \cap (0, \infty) \neq \emptyset, \quad (A \cap (-\infty, 0)) \cup (A \cap (0, \infty)) = A$$

### Definition 1.4.3: Path

A **path** in a given subset  $A \subseteq \mathbb{R}^n$  is a continuous map

$$\gamma : [0, 1] \rightarrow \mathbb{R}^n \quad \text{with } \gamma([0, 1]) \subseteq A$$

### Definition 1.4.4: Path connected subset

A subset  $A \subseteq \mathbb{R}^n$  is **path connected** if for any two elements  $x, y \in A$  there exists

$$\gamma : [0, 1] \rightarrow A \quad \text{s.t. } \gamma(0) = x, \gamma(1) = y$$

### Proposition 1.4.1 Connectedness and path connectedness

- 1) For  $n = 1$ , a subset  $A \subseteq \mathbb{R}$  is connected iff it is path connected, iff it is an interval.
- 2) Every path connected subset is connected.
- 3) A nonempty open subset  $U \subseteq \mathbb{R}^n$  is connected iff it is path connected.
- 4) There are examples of connected subsets  $A$  that are not path connected. (HW)

## 1.5 Compactness

### Definition 1.5.1: Open cover

A collection  $\{U_\alpha \mid \alpha \in I\}$  of open subsets of  $\mathbb{R}^n$  is an **open cover** of a subset  $A \subseteq \mathbb{R}^n$  if

$$A \subseteq \bigcup_{\alpha \in I} U_\alpha$$

$\mathbb{R}^n$  is covered by the collection of  $\epsilon$ -balls  $B_\epsilon(x)$  for any given  $\epsilon > 0$ .

### Definition 1.5.2: Compact set

A subset  $C \subseteq \mathbb{R}^n$  is **compact** if every open cover  $\{U_\alpha \mid \alpha \in I\}$  of  $C$  contains a finite subcover: there exist  $\alpha_1, \dots, \alpha_k \in I$  such that

$$C \subseteq \bigcup_{i=1}^k U_{\alpha_i}$$

### Definition 1.5.3: Boundedness

A subset  $C \subseteq \mathbb{R}^n$  is called **bounded** if it is contained in  $B_R(0)$  for  $R > 0$  sufficiently large.